📊 RURO Post-Estimation Analysis

Generated: 2026-06-01 09:32:37

Total Parameters: 50 | **Groups:** joint

Estimation Results Source: `C:\Users\hisham\MNL\EUROMOD-STORAGE\outputs\post_estimation\realdata_joint_901_gsplit_UNIFIED\joint\estimation_results.json`

Estimation Configuration

Specification	joint_pooled_v1_bll0_tlmpin_gsplit
Specification File	`scripts\bpool\specs\estimation_spec_joint_pooled_v1_bll0_tlmpin_gsplit.yaml`
Model Family	regular
Proposal Correction	Enabled
Opportunity Centering	Enabled

Single Males Single Females Males in Couples Females in Couples

⏱ Elapsed Time

Estimation	Post-Estimation	Total	Iterations
24m 23s	**1m 58s**	**26m 22s**	**2352**

⚙ Solver & Convergence Diagnostics (from DiagnosticsBundle)

Solver	L-BFGS-B (scipy, box) -> optimistix BFGS polish (JAX) (family: bfgs)
Objective (log-likelihood)	-238362.7881
Wall time (s)	1463.76
Per-group convergence:	

group	success	message	iters	nfev	grad_norm	wall(s)
joint	True	L-BFGS-B (scipy, box) -> optimistix BFGS polish (JAX)	2352	2570	32.605994	1463.76

📈 A. Core Likelihood and Sample Statistics

Log-likelihood	-238362.7881
Observations (rows)	12445
Choice sets / groups	12445
Alternatives per choice set	1.00
Free parameters	50
Fixed parameters	0
AIC	476825.5763
BIC	477197.0300
AIC / n_obs	38.314630

📈 B. Null-Model and Pseudo-R^2 Diagnostics

ll_null (uniform)	-73712.3805
ll_null (prior-corrected)	-263173.1206
ρ^2 (uniform)	-2.2337
ρ^2 (prior-corrected)	0.0943
Adj. ρ^2 (uniform)	-2.2344
Adj. ρ^2 (prior-corrected)	0.0941

ρ^2 values use McFadden's formulation 1 - LL/LL0. For sampled-alternative / job-choice models the prior-corrected null is the right comparison; the uniform null is kept for legacy comparability.

📈 C. Bound and Fixed-Parameter Diagnostics

Parameters (total)	50
Free parameters	50
Fixed parameters	0
Parameters with bounds	49
At lower bound	1
At upper bound	2

Parameters at or near bounds

parameter	side	estimate	bound	distance
joint.beta_l_age2_sf	upper	1.000000	1.000000	0.000000e+00
joint.beta_l0_m	lower	1.000000e-06	1.000000e-06	0.000000e+00
joint.beta_l_age2_f	upper	1.000000	1.000000	0.000000e+00

📈 D. Economic Sanity Diagnostics

Not available: Marginal-utility diagnostics not computed (requires --mnl-base).

▶ 📈 **Legacy combined fit-statistics dump (technical appendix)**

🔍 Identification & Hessian Diagnostics (from DiagnosticsBundle)

Condition number κ	1.45e+06

Min eigenvalue	0.4080
Max eigenvalue	—
Negative eigenvalues	0

🔍 Identification Diagnostics

Hessian-based diagnostics for parameter identification quality.

Condition Number: 1.45e+06

✗ Severe Identification Problem

The Hessian is severely ill-conditioned (κ ≥ 10,000). This indicates serious identification issues with unreliable parameter estimates.

Eigenvalue Analysis

Minimum Eigenvalue

4.08e-01

Maximum Eigenvalue

N/A

Negative Eigenvalues

0

✓ Local maximum confirmed

ℹ️ Technical Notes

Condition Number (κ): Measures sensitivity of parameter estimates to small changes in data. $\kappa = \lambda_{max} / \lambda_{min}$, where λ are eigenvalues of -H (Hessian).

Eigenvalues: All eigenvalues of -H should be positive for a local maximum. Near-zero eigenvalues indicate flat directions (weak identification).

Sign convention: Eigenvalues shown come from the Hessian used to compute SEs. When SEs are computed numerically, this is the Hessian of the negative log-likelihood (H), so negative values indicate H is not positive semidefinite (ill-conditioning or non-optimum).

Standard Errors: Computed as $SE = \sqrt{\mathrm{diag}((-H)^{-1})}$, where H is the Hessian at the optimum. Large SE indicates parameter is not precisely estimated.

📐 Model Specification

Utility and opportunity functions for each demographic group, shown in both symbolic and numerical form.

📐 Model Index

$$V_{ij} = U_{ij} + O^{E}_{ij} + O^{H}_{ij} + O^{W}_{ij} + O^{Occ}_{ij} + - \log \text{prior}_{ij}$$

$$P_{ij} = \exp(V_{ij}) \;/\; \Sigma_{k \in C_i} \exp(V_{ik})$$

U is the utility of consumption and leisure; the $O^{\cdot}$ terms are opportunity-density contributions (employment/hours, wage, occupation); `log prior` rescales each alternative to the data-generating measure. The terms displayed reflect the opportunity blocks declared in the active YAML specification.

⏰ Employment and Hours Opportunity Parameters

Working-gated employment and hours intercepts. Combines `hours_opportunity.shifters` (base employment intercept and PT/FT focal-point indicators) with `market_opportunity.shifters` (working-gated residual market access terms like gsur, education).

Symbolic form

$$O^{E} + O^{H} =$$

```
      beta_E * working
+ beta_h_pt1 * working_pt1
+ beta_h_pt2 * working_pt2
+ beta_h_ft * working_ft
+ beta_h_lh * working_lh
+ beta_E_gsur * gsur * working
```

```
+ beta_E_drgn2 * reg2 * working
+ beta_E_drgn3 * reg3 * working
+ beta_E_drgn4 * reg4 * working
+ beta_E_drgn5 * reg5 * working
+ beta_E_drgn6 * reg6 * working
+ beta_E_drgn7 * reg7 * working
+ beta_E_drgn8 * reg8 * working
+ beta_E_y2015 * year_2015_indicator * working
+ beta_E_y2017 * year_2017_indicator * working
+ beta_E_drgur * drgur * working
+ beta_E_drgmd * drgmd * working
```

Numerical form (by group)

Joint (All Groups): -0.3311 * working + -1.4275 * working_pt1 + -1.2108 * working_pt2 + 1.0338 * working_ft + -1.2443 * working_lh + -1.4041 * gsur * working + -0.0701 * reg2 * working + 0.0845 * reg3 * working + 0.0027 * reg4 * working + -0.1415 * reg5 * working + -0.0538 * reg6 * working + -0.1811 * reg7 * working + -0.3444 * reg8 * working + -0.2510 * year_2015_indicator * working + -0.0642 * year_2017_indicator * working + -0.5323 * drgur * working + -0.6644 * drgmd * working

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\log(wage) = \mu_w(X) + \varepsilon, \varepsilon \sim N(0, \sigma^2)$. The mean μ_w is built from `wage_opportunity.mean_shifters`.

Symbolic form

```
μw =

      beta_w0
+ beta_w_educL · educL
+ beta_w_educH · educH
+ beta_w_pexp · pexp_years
+ beta_w_pexp2 · pexp_years2
```

```
log(wage) = μ_w + ε,    ε ~ N(0, sigma²)
```

Numerical form (by group)

Joint (All Groups): μ_w = 2.1971 + -0.0606 * educL + 0.3384 * educH + 0.3816 * pexp_years + -0.0818 * pexp_years2

sigma = 0.3899

Occupation Opportunity Parameters

Working-gated occupation shifters. Each `applies_to` group has its own coefficient set on the non-reference categories of `loc4`. Reference category: `loc4 = 1`.

male (`applies_to = male`)

O^{Occ}_{male} =

```
    beta_occ_2_m * loc4_2 * working
+ beta_occ_3_m * loc4_3 * working
+ beta_occ_4_m * loc4_4 * working
```

-1.6265 * loc4_2 * working + -2.3279 * loc4_3 * working + 0.2572 * loc4_4 * working

female (`applies_to = female`)

O^{Occ}_{female} =

```
    beta_occ_2_f * loc4_2 * working
+ beta_occ_3_f * loc4_3 * working
+ beta_occ_4_f * loc4_4 * working
```

```
            0.0179 * loc4_2 * working  +  -0.4041 * loc4_3 * working  +
0.8376 * loc4_4 * working
```

📈 Curvature-Based Heuristics (Structural Elasticity Approximations)

⚠️ Interpretation Note

These are **not** true labor supply elasticities. They are heuristic approximations derived from the curvature parameters (θ) of the Box-Cox utility function. The Hicksian approximation is (1 - θ_l), which measures the curvature of preferences. For rigorous elasticity estimates, use simulation-based methods that account for the full discrete choice structure and budget constraints.

📊 Labor Supply Elasticities

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Males	2.790	2.690	0.837	1.953	-1.790	0.010	4.429	1.000
Single Females	2.440	2.340	0.732	1.708	-1.440	0.010	3.848	1.000
Males in Couples	1.800	1.700	0.540	1.260	-0.800	0.010	0.000	1.000
Females in Couples	3.227	3.127	0.968	2.259	-2.227	0.010	10.259	1.000

🎯 Fit Diagnostics

Observed vs Predicted

Group	Obs. Participation	Pred. Participation	Obs. Mean Hours	Pred. Mean Hours
Single Males	91.9%	99.8%	39.4	39.9
Single Females	93.0%	99.7%	36.1	39.5
Males in Couples	97.2%	99.5%	41.5	38.2
Females in Couples	96.2%	99.5%	35.6	38.0

⚠️ Marginal Utility Diagnostics

Analysis at chosen alternatives. Negative values indicate potential specification issues.

Negative MUC/MUL Summary

MUC Behavior Analysis

For well-behaved utility: MUC > 0 ($\beta_c > 0$) and diminishing ($\theta_c < 1$)

Group	N Individuals	% Neg MUC	% Neg MUL	MUC Mean	MUL Mean
Single Males	2764	0.00%	0.00%	0.0239	2.2082e-04
Single Females	2243	0.00%	0.00%	0.0245	9.5360e-04
Males in Couples	7438	0.00%	0.00%	0.0160	1.9109e-09
Females in Couples	7438	0.00%	0.00%	0.0160	1.0324e-04

Group	β_c	θ_c	MUC Positive?	MUC Diminishing?	Well-Behaved?	MUC at Median C	C where MUC=1	Notes
Single Males	1.0000	0.5000	✓	✓	✓	1.0000	1.0000	
Single Females	1.0000	0.5000	✓	✓	✓	1.0000	1.0000	
Males in Couples	1.0000	0.0000	✓	✓	✓	1.0000	1.0000	
Females in Couples	1.0000	0.0000	✓	✓	✓	1.0000	1.0000	

📊 Choice Probability Diagnostics

Probability Sum Sanity Check

Max |Σp - 1|	0.000000
Mean |Σp - 1|	0.000000
% HH off by >0.01	0.00%
% HH off by >0.001	0.00%

P(chosen) Distribution

Min	0.0000
10th percentile	0.0000
25th percentile	0.0000
Median	0.0000
Mean	0.0000
75th percentile	0.0000
90th percentile	0.0000
Max	0.0001

📊 Probability & Fit Diagnostics (from DiagnosticsBundle)

Probability-sum sanity check

max_error	4.440892e-16
mean_error	7.157348e-17
pct_off_by_0.01	0.000000
pct_off_by_0.001	0.000000

P(chosen) distribution

min	2.443980e-20
max	1.044884e-04
mean	2.101438e-06
median	5.446513e-10

q10	3.507928e-12
q25	3.039884e-11
q75	1.717371e-06
q90	5.523578e-06

Worst-fit households (top 10)

#	idhh	group	p_chosen	ll_i
1	200002989700	cou	2.443980e-20	-45.1581
2	300003319200	cou	7.968047e-20	-43.9763
3	200001966500	cou	1.208849e-19	-43.5594
4	100001229900	cou	2.654646e-19	-42.7728
5	200004300600	cou	3.015515e-19	-42.6453
6	100000054000	cou	4.365518e-19	-42.2754
7	200004241200	cou	4.869634e-19	-42.1661
8	200004264600	cou	1.764170e-18	-40.8789
9	200001729600	cou	2.369626e-18	-40.5838
10	300001899000	cou	4.216289e-18	-40.0076

⚠️ Parameters at Bounds (within 1e-6)

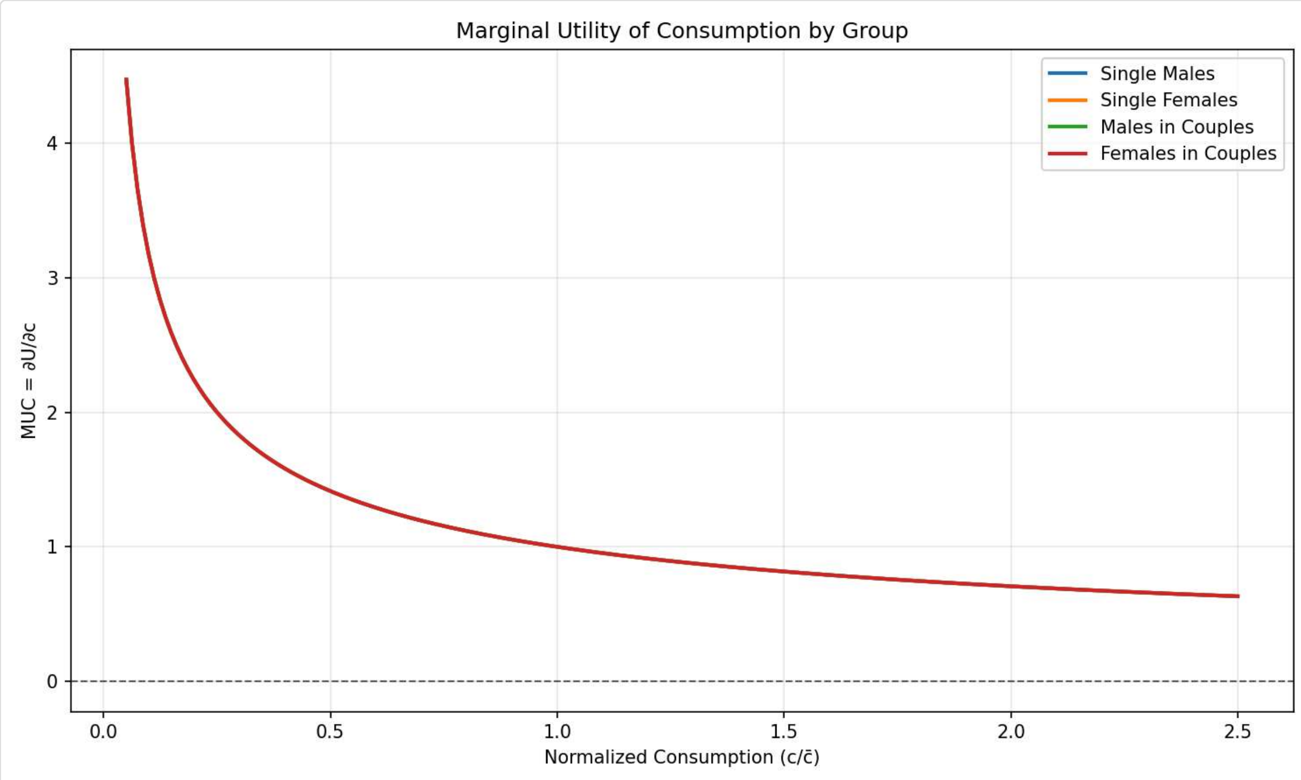
These parameters hit their constraints. Consider checking if bounds are too restrictive.

Parameter	Estimate	Bound	Side
joint.beta_l_age2_sf	1.000000	1.000000	⬆️ upper
joint.beta_l0_m	0.000001	0.000001	⬇️ lower
joint.beta_l_age2_f	1.000000	1.000000	⬆️ upper

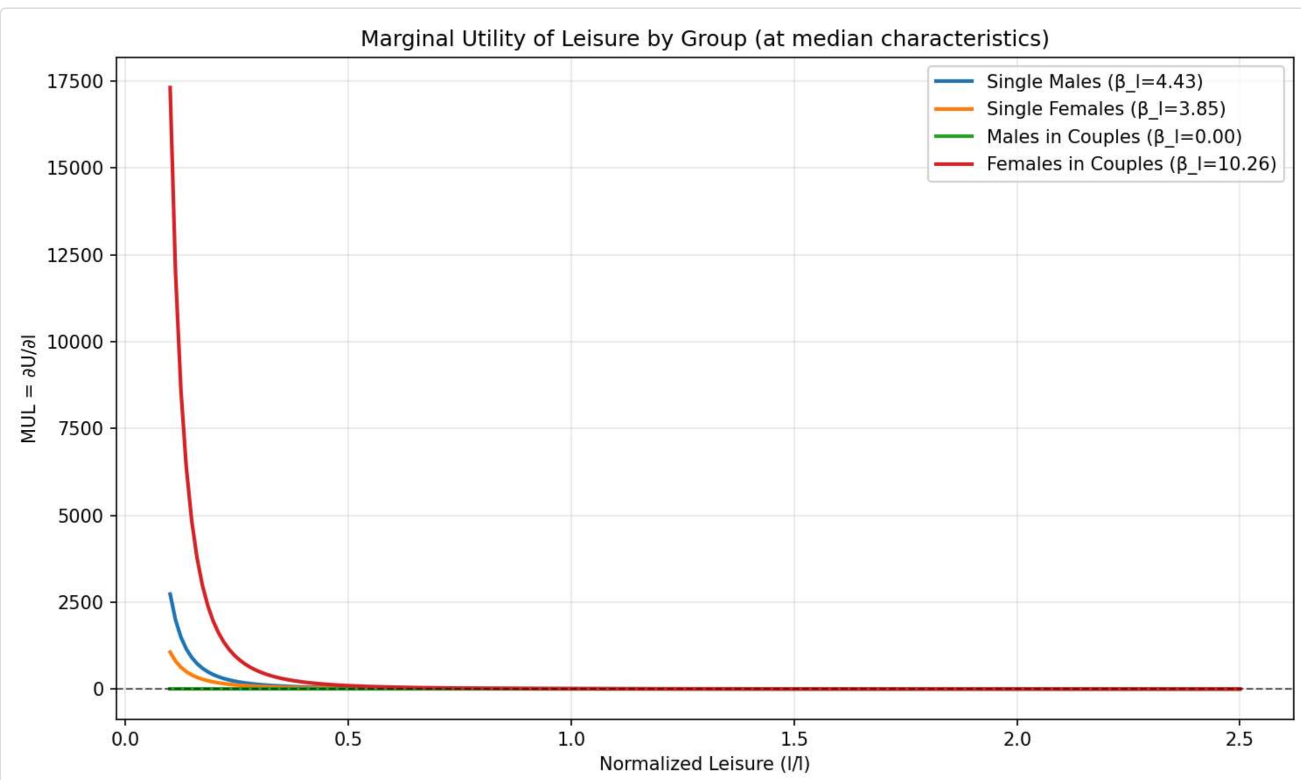
🗺️ Utility Contours & Plots

Marginal Utility Comparison

Marginal Utility of Consumption

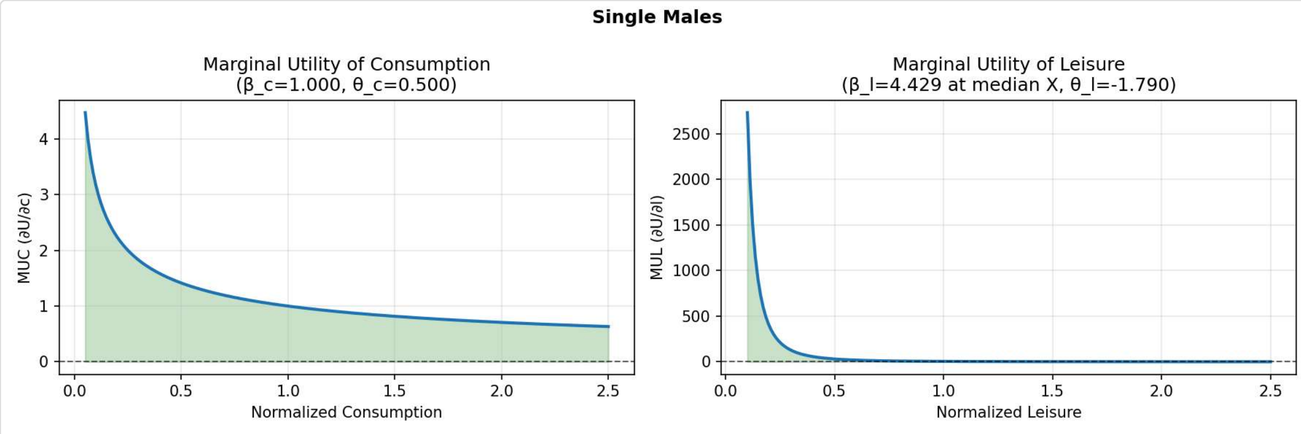


Marginal Utility of Leisure

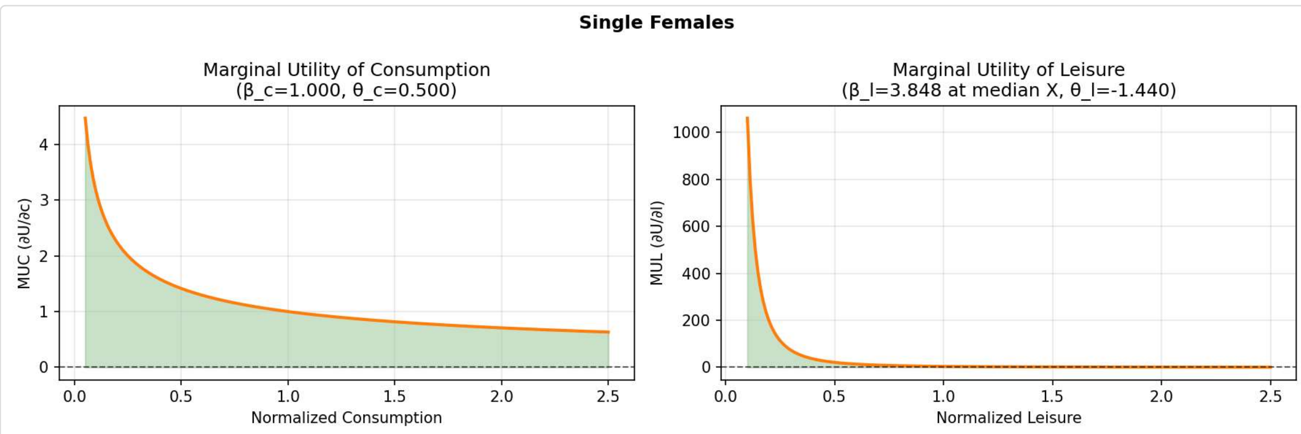


📊 Marginal Utility Distributions by Group

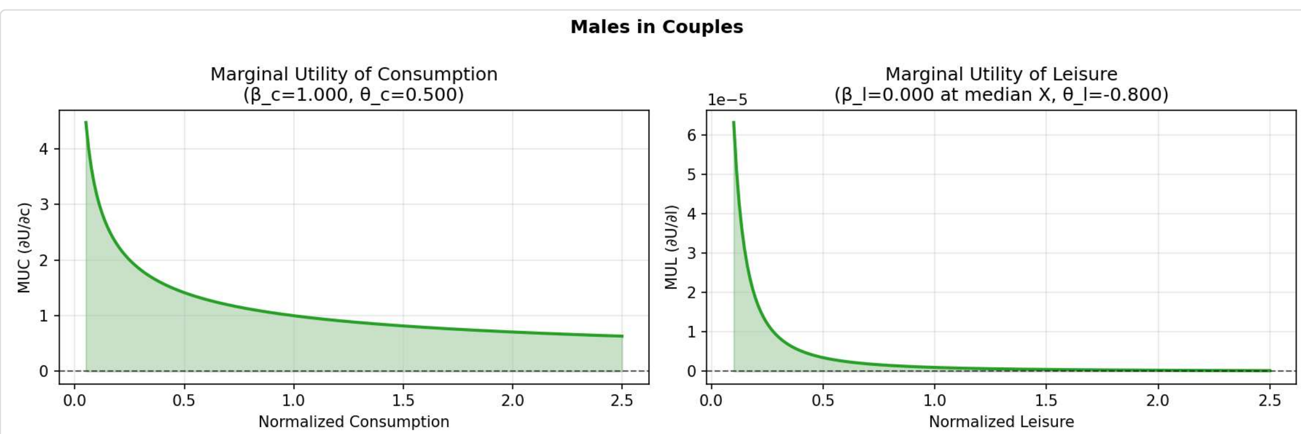
Single Males - MUC & MUL Distributions



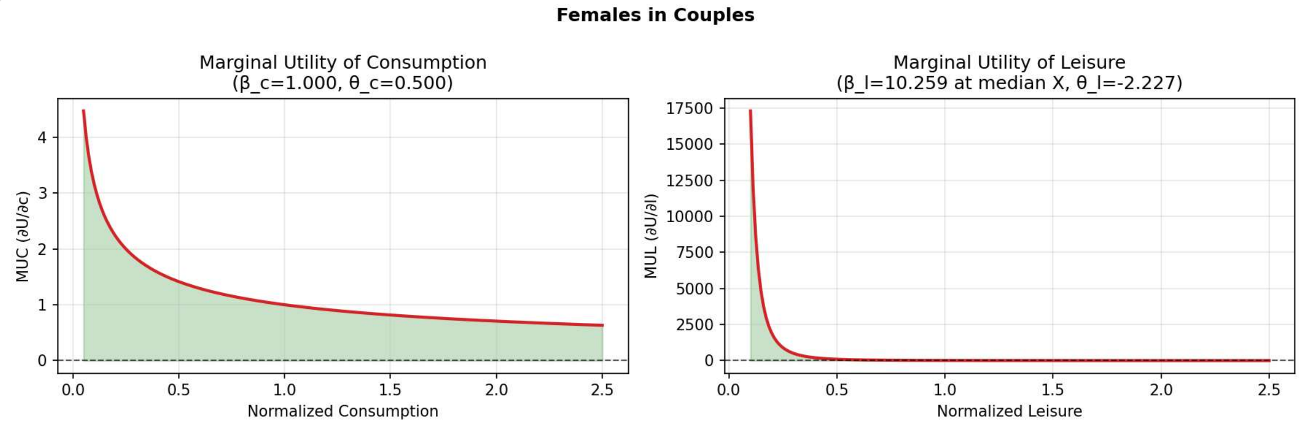
Single Females - MUC & MUL Distributions



Males in Couples - MUC & MUL Distributions

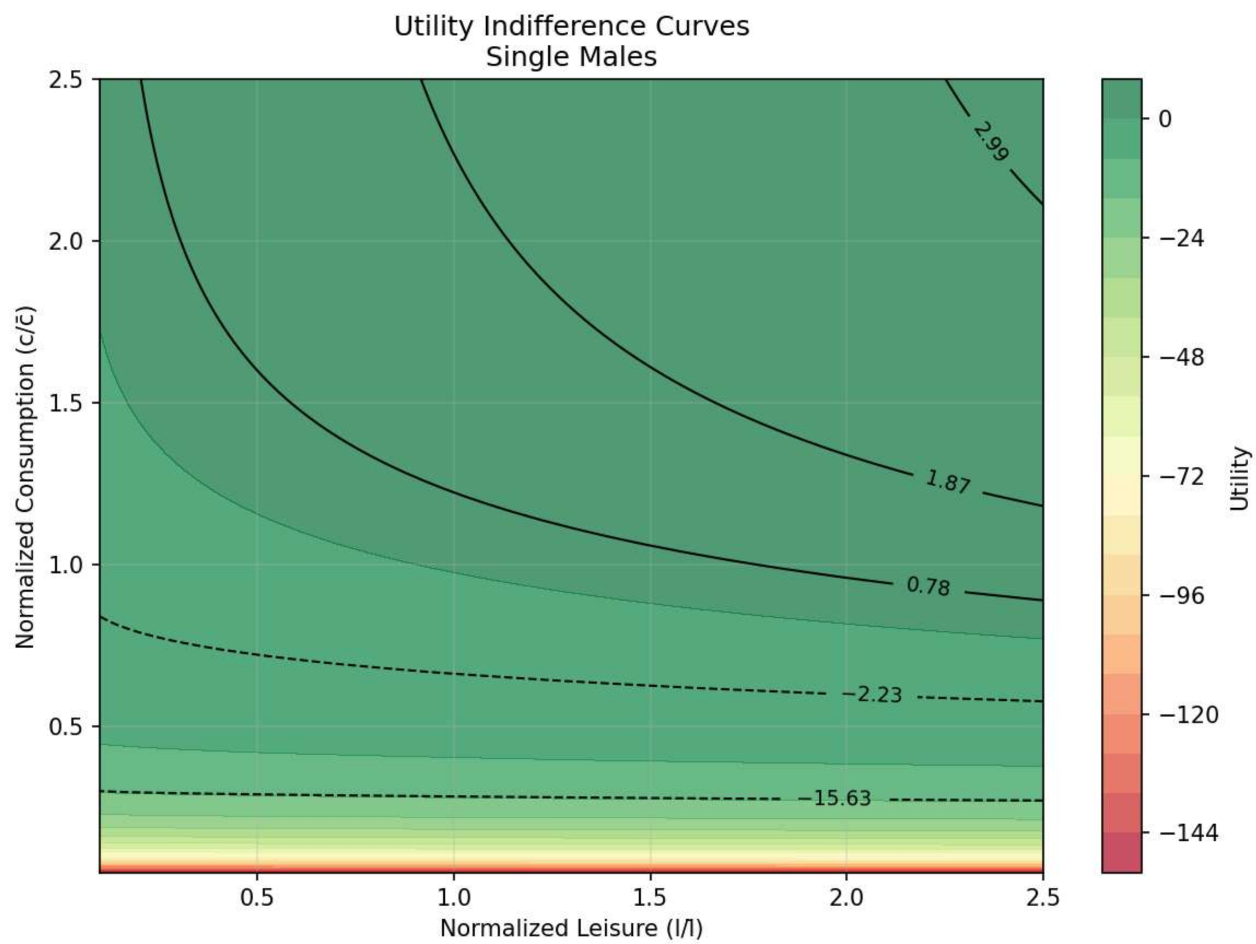


Females in Couples - MUC & MUL Distributions

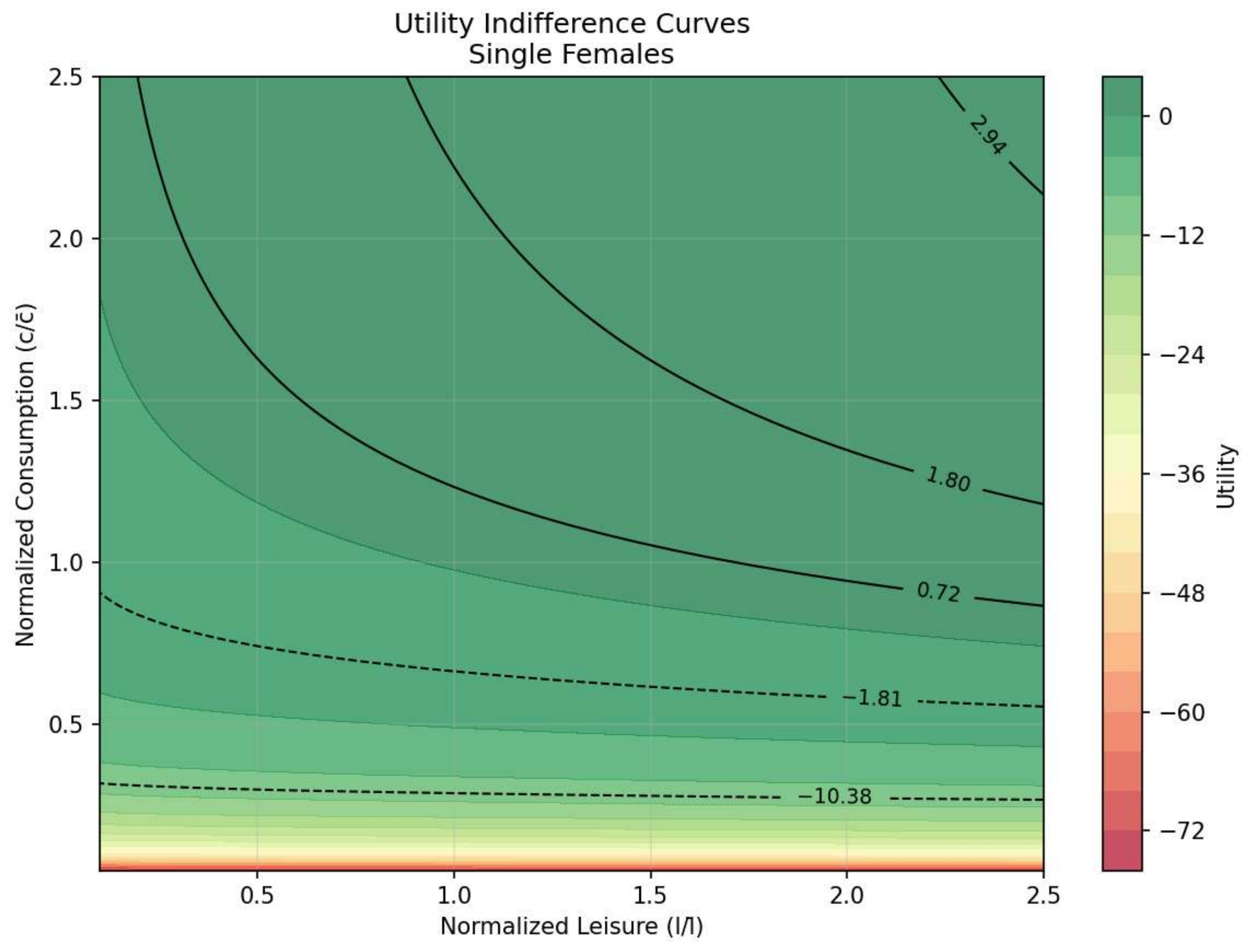


Utility Indifference Curves by Group

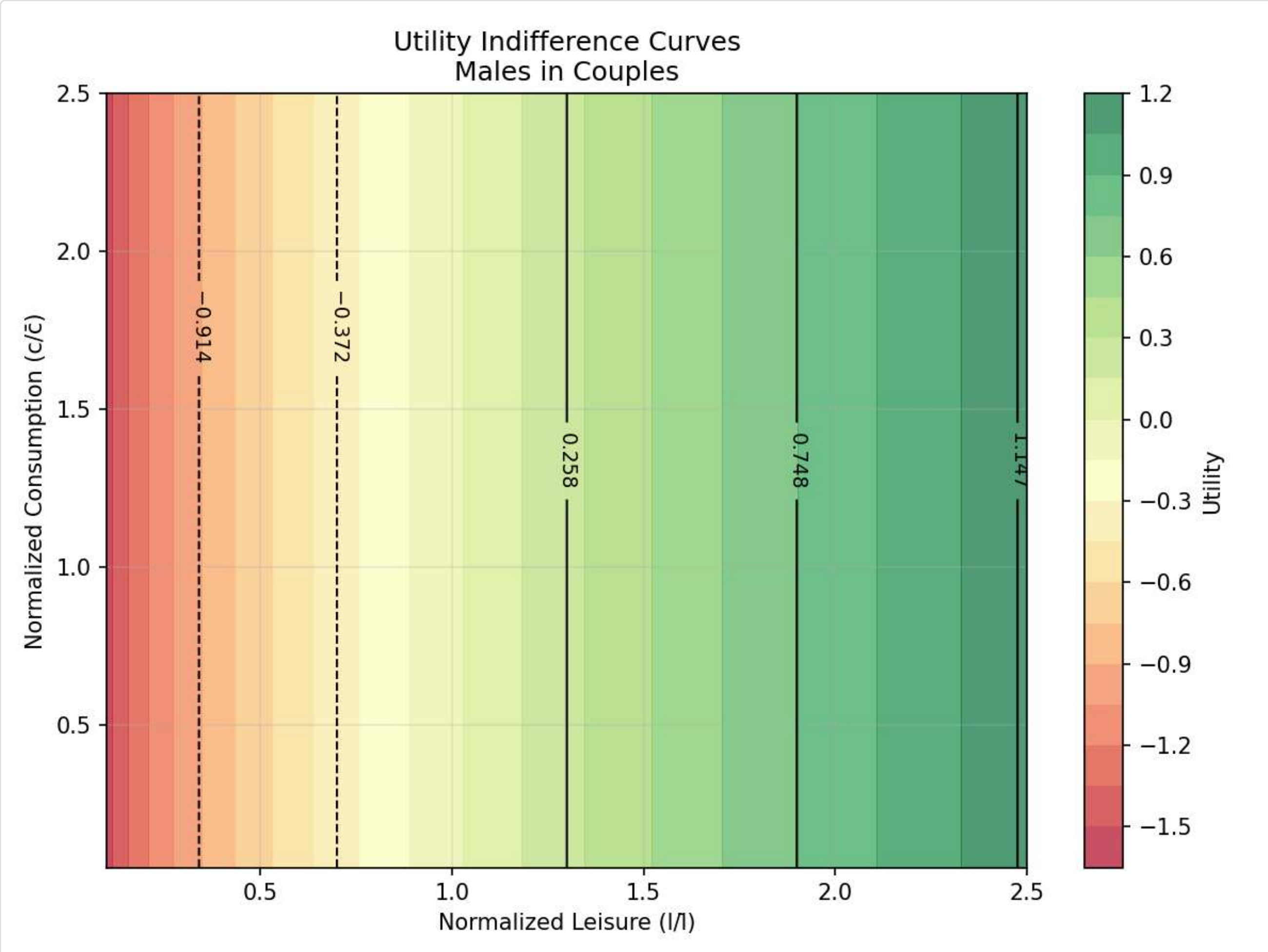
Single Males



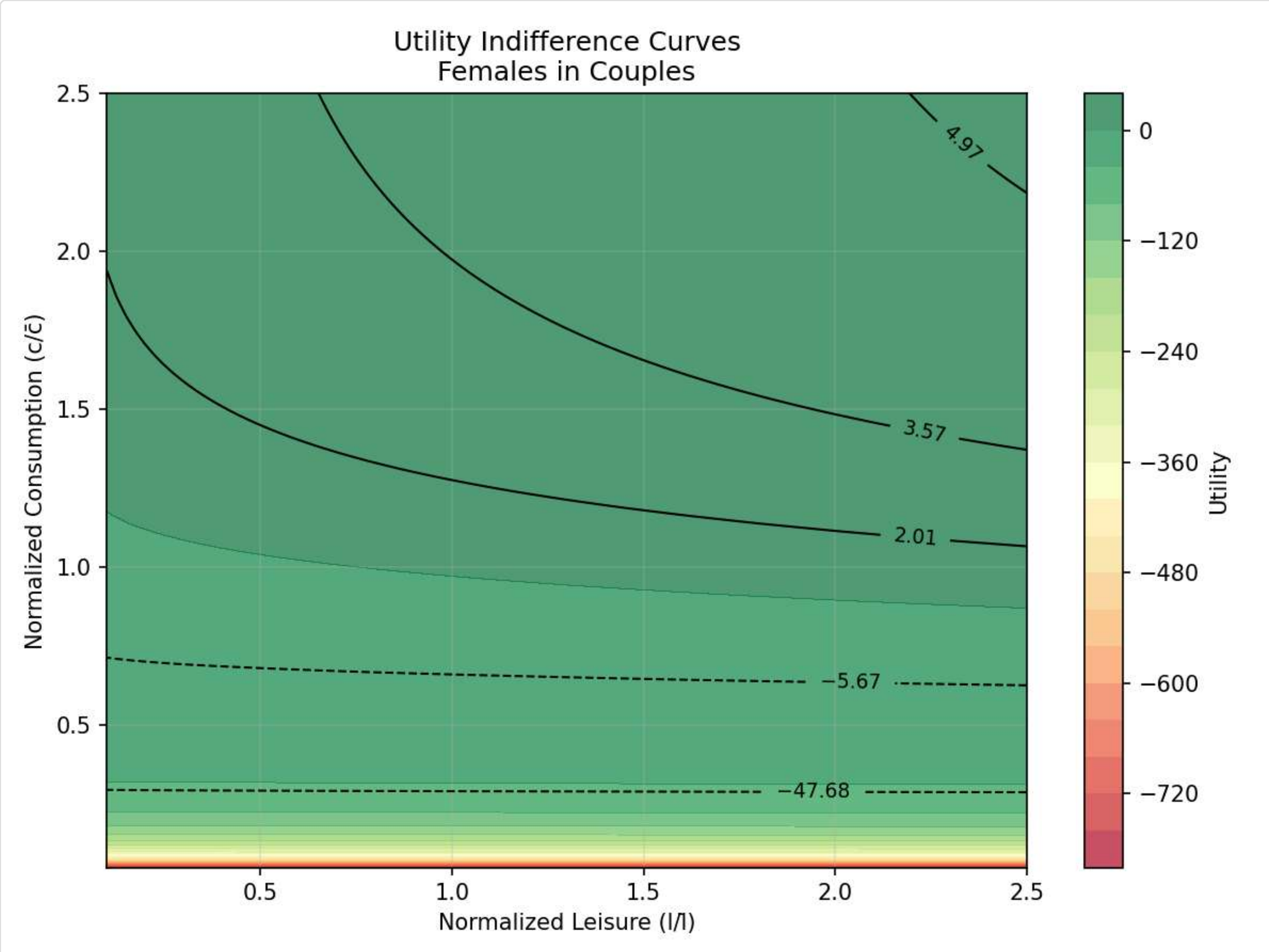
Single Females



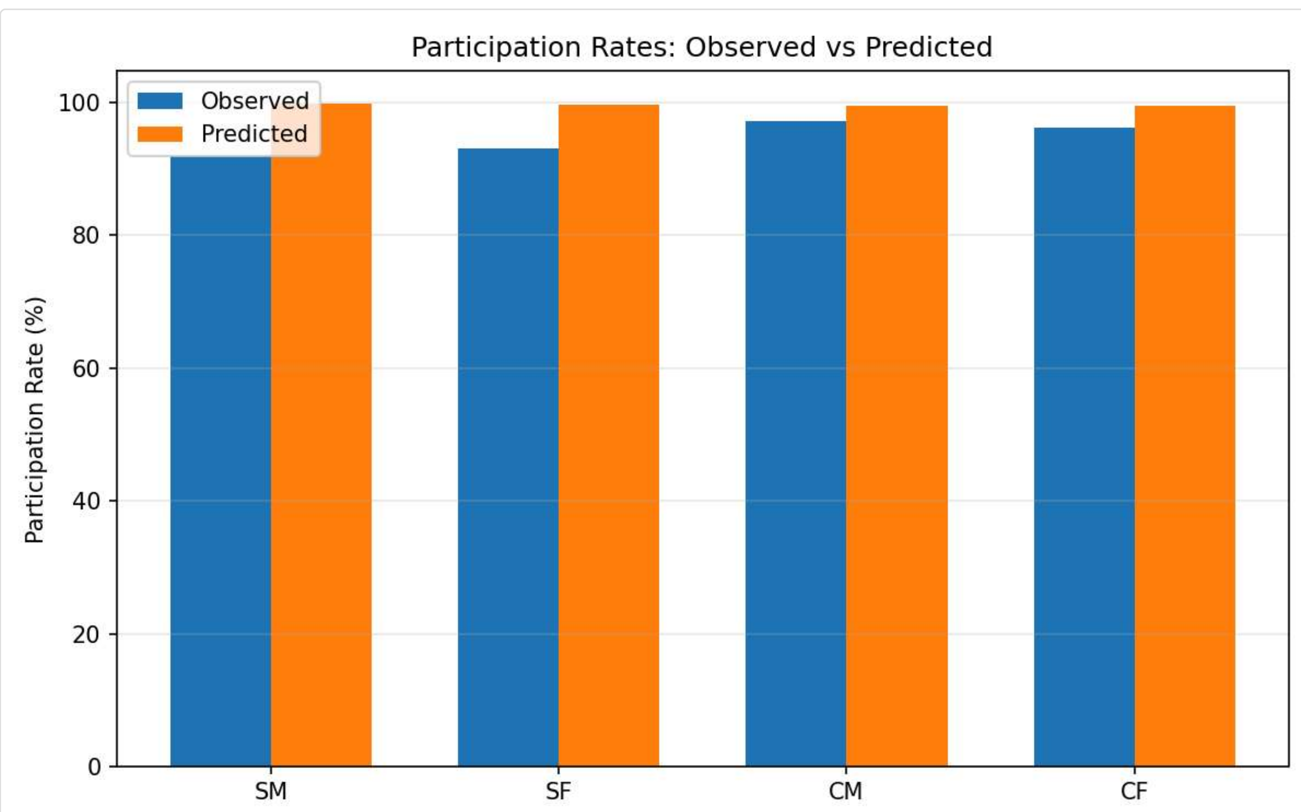
Males in Couples



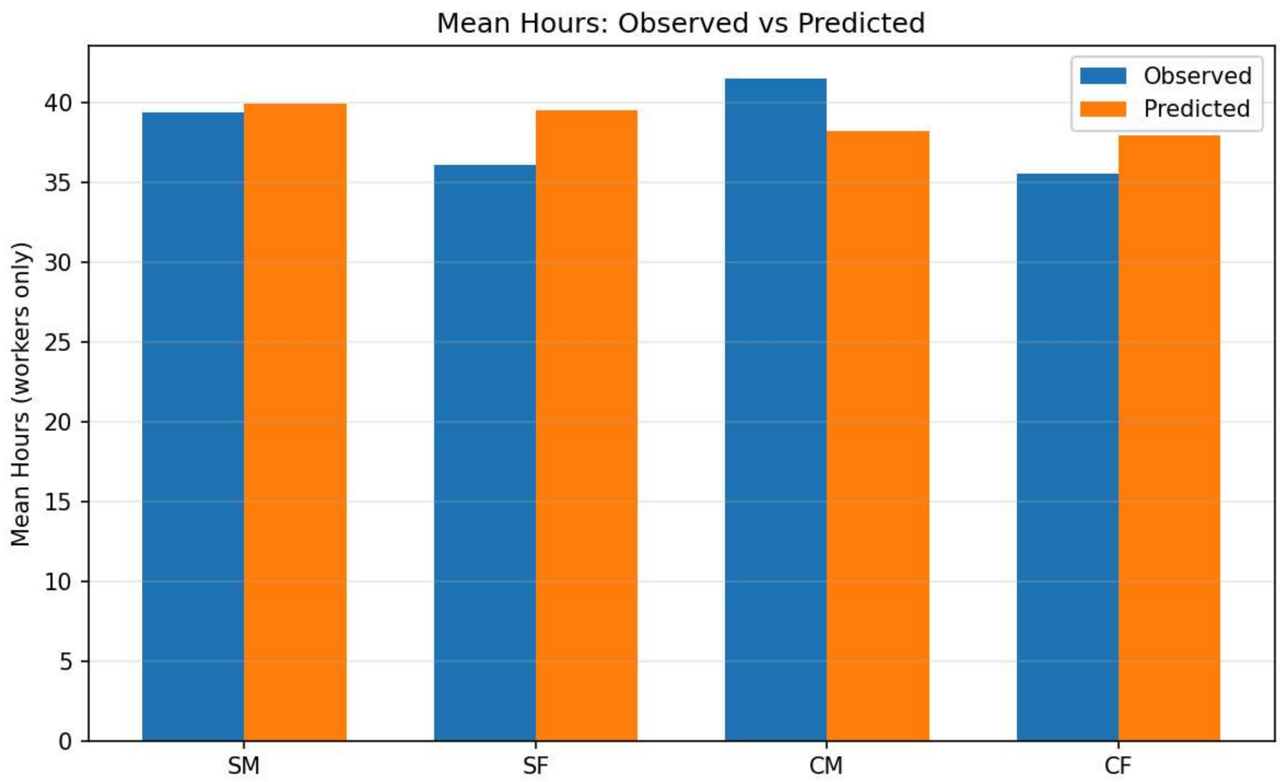
Females in Couples



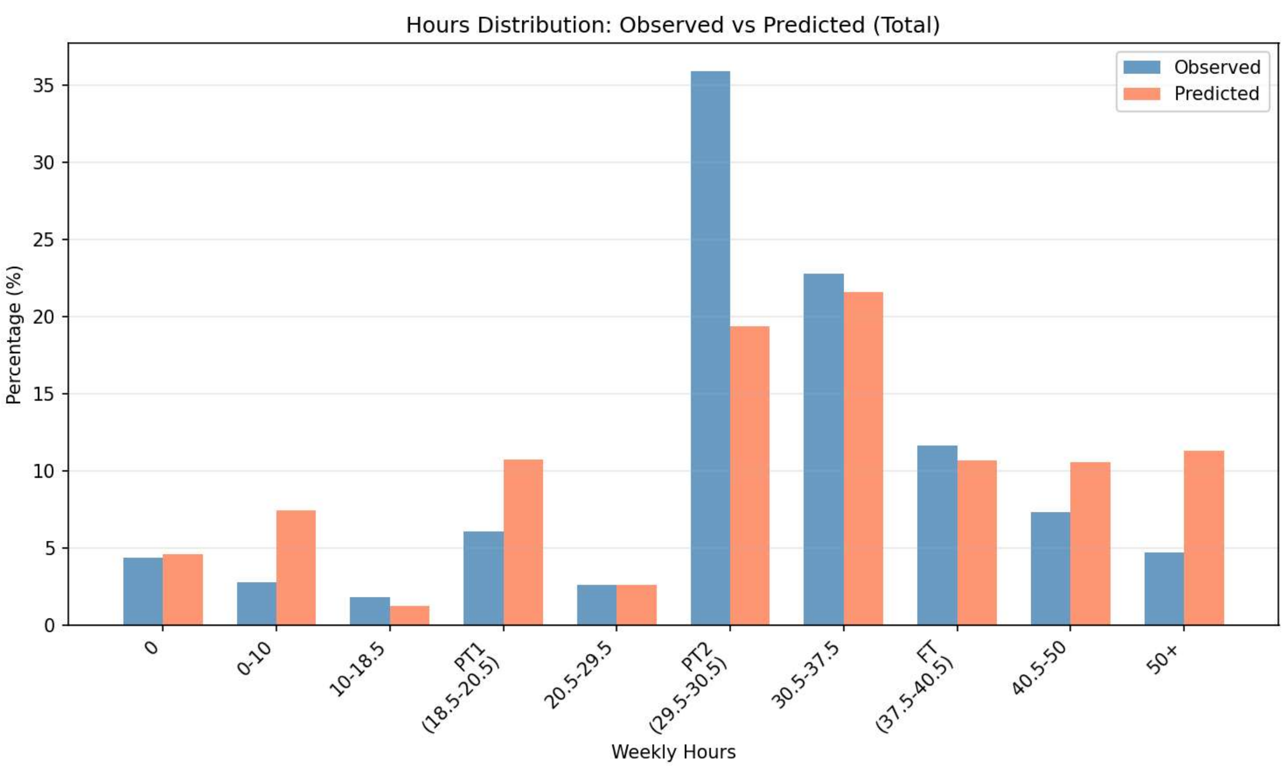
Participation

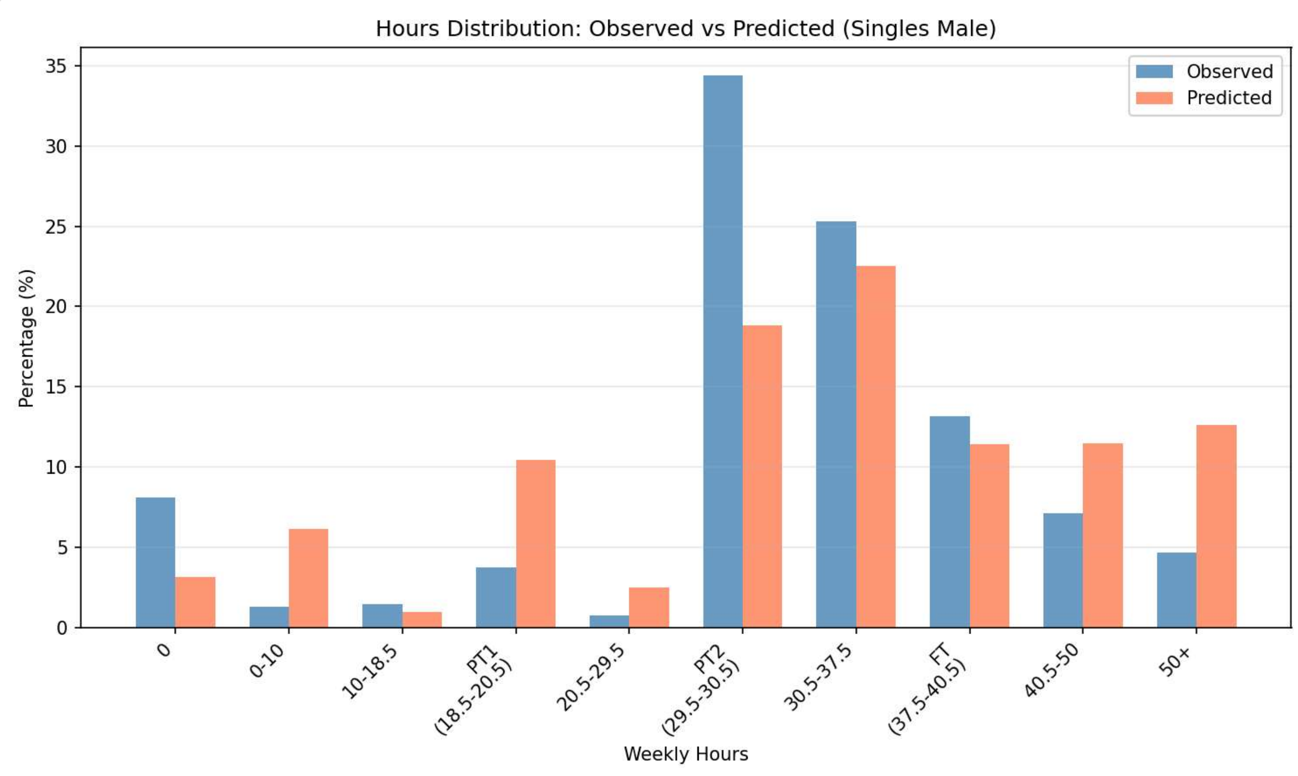


Mean Hours

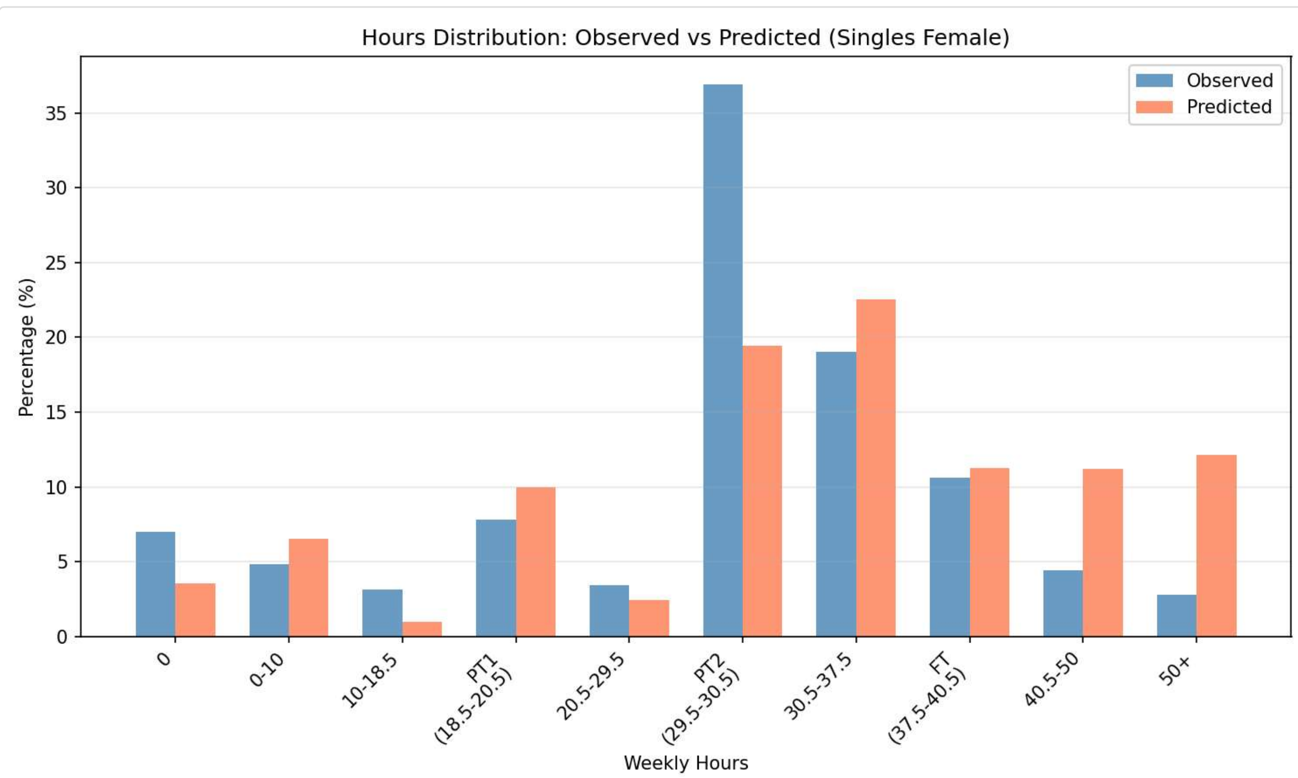


?? Hours Distribution: Observed vs Predicted

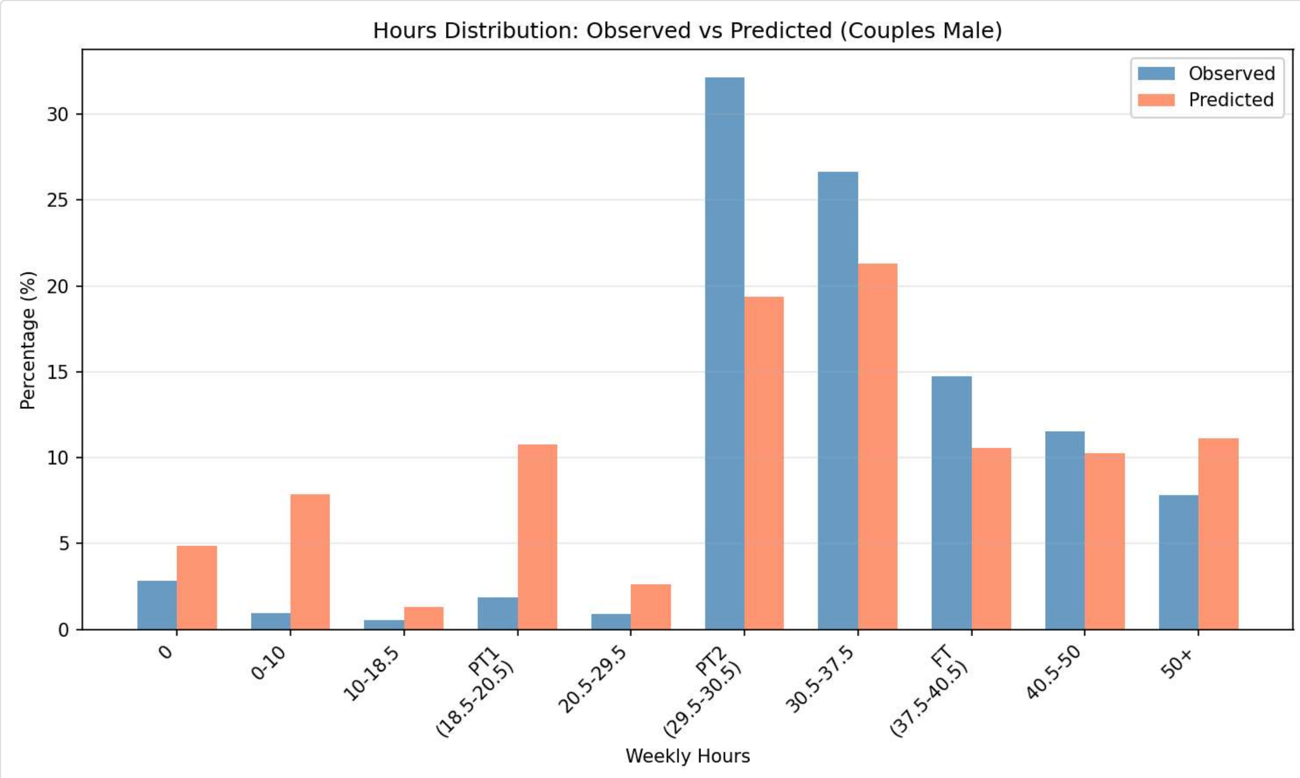


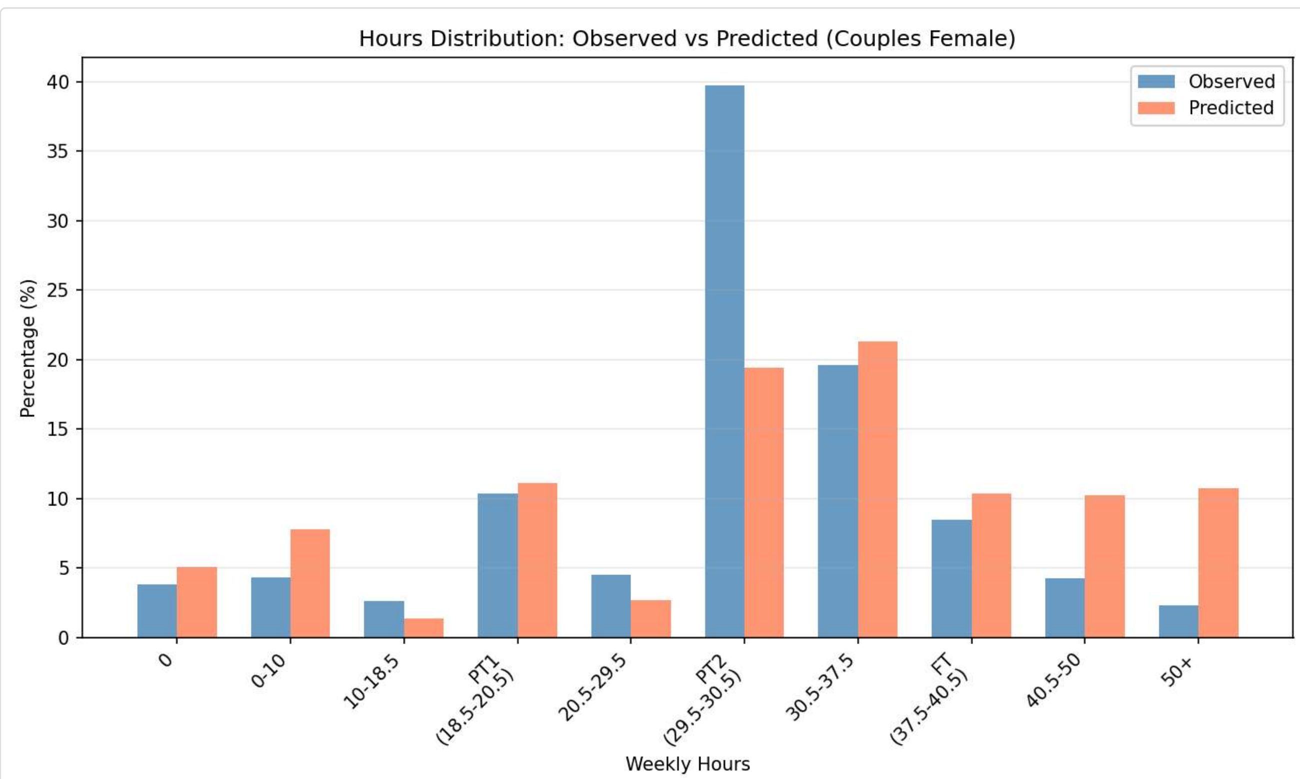


Hours Distribution: Observed vs Predicted (Singles Male)
Observed
Predicted
Percentage (%)
0
5
10
15
20
25
30
35
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+
Weekly Hours

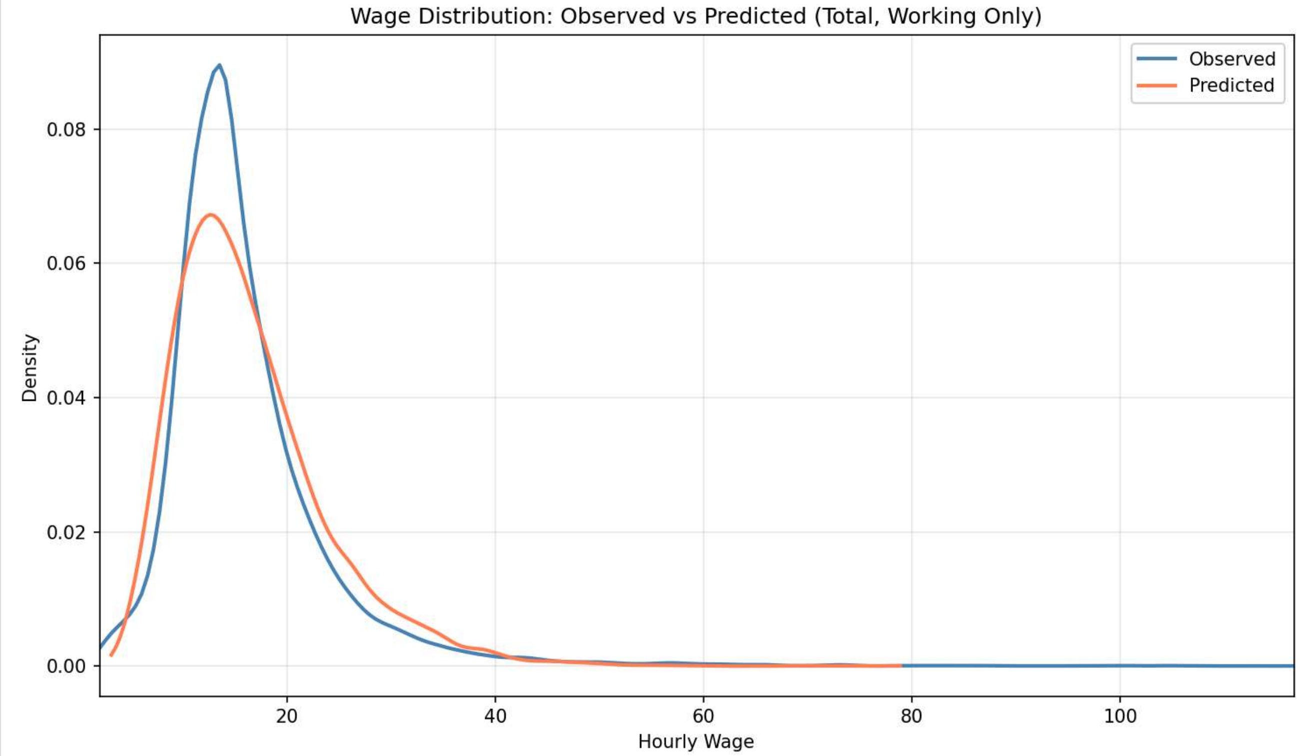


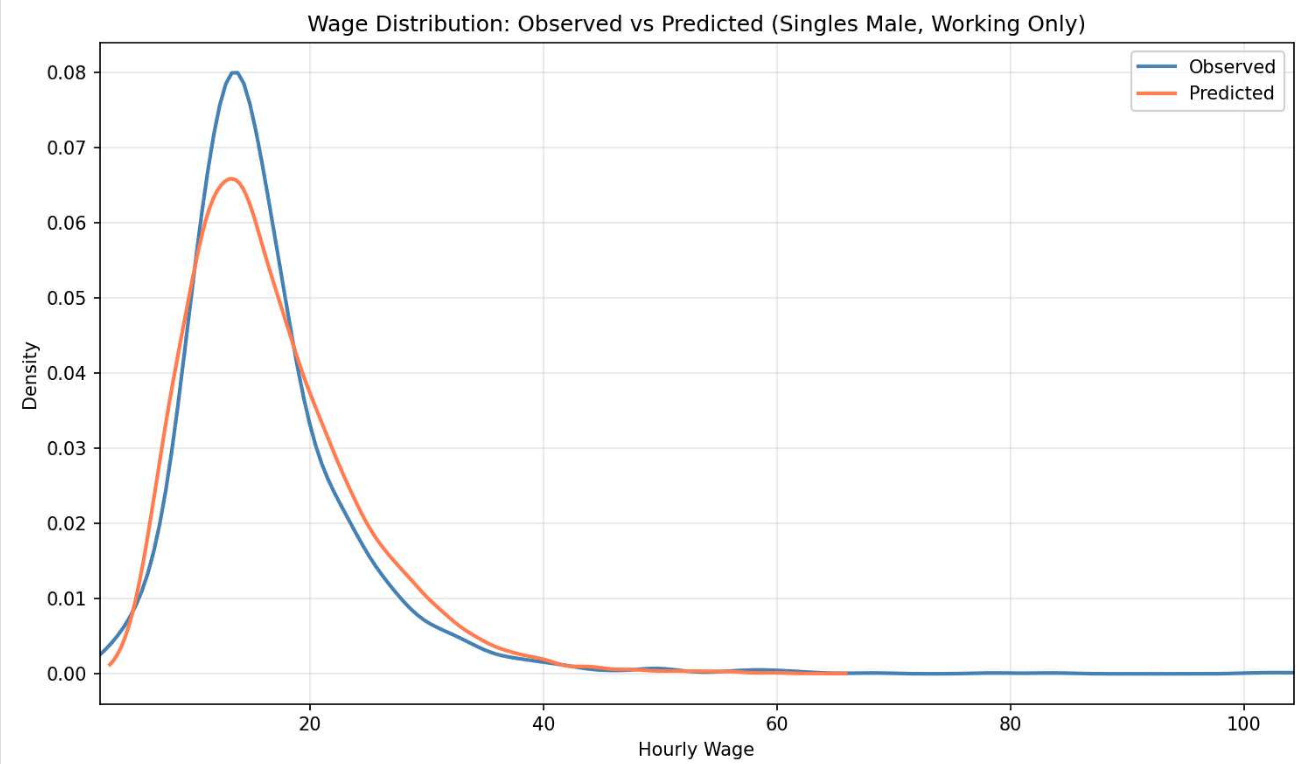
Hours Distribution: Observed vs Predicted (Singles Female)
Observed
Predicted
Percentage (%)
0
5
10
15
20
25
30
35
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+
Weekly Hours

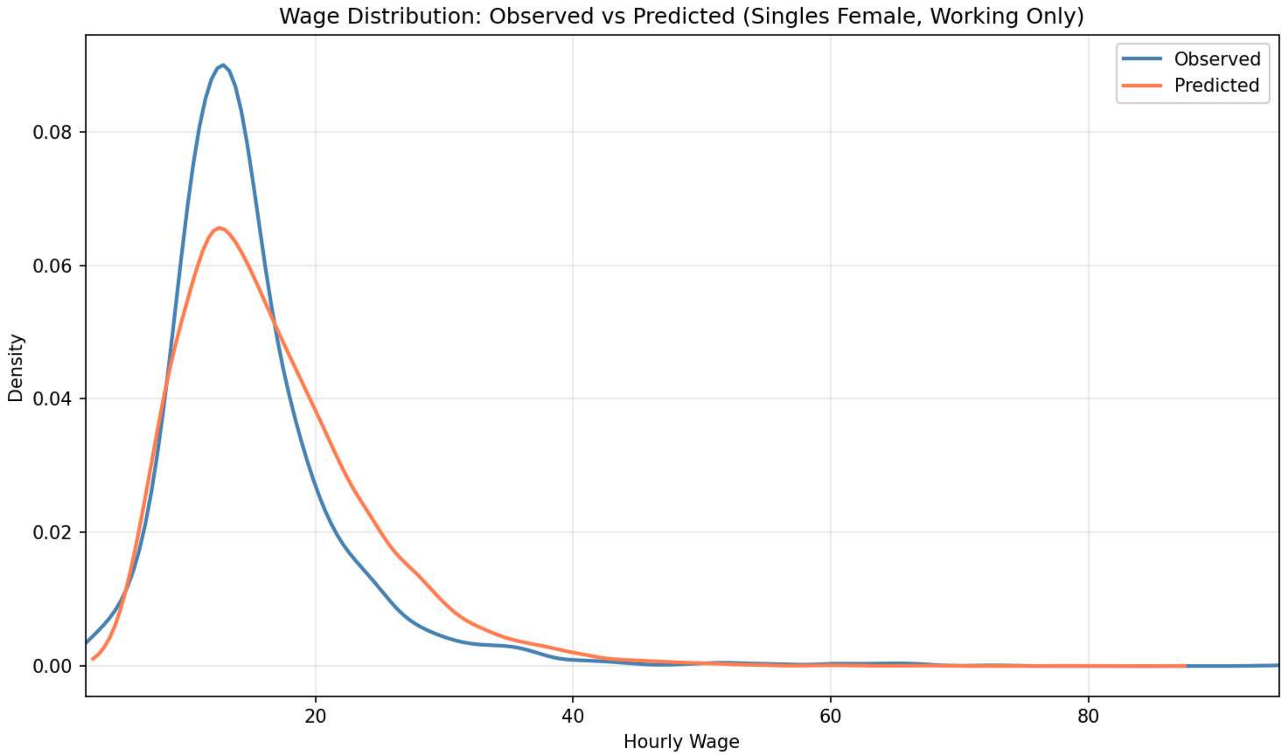


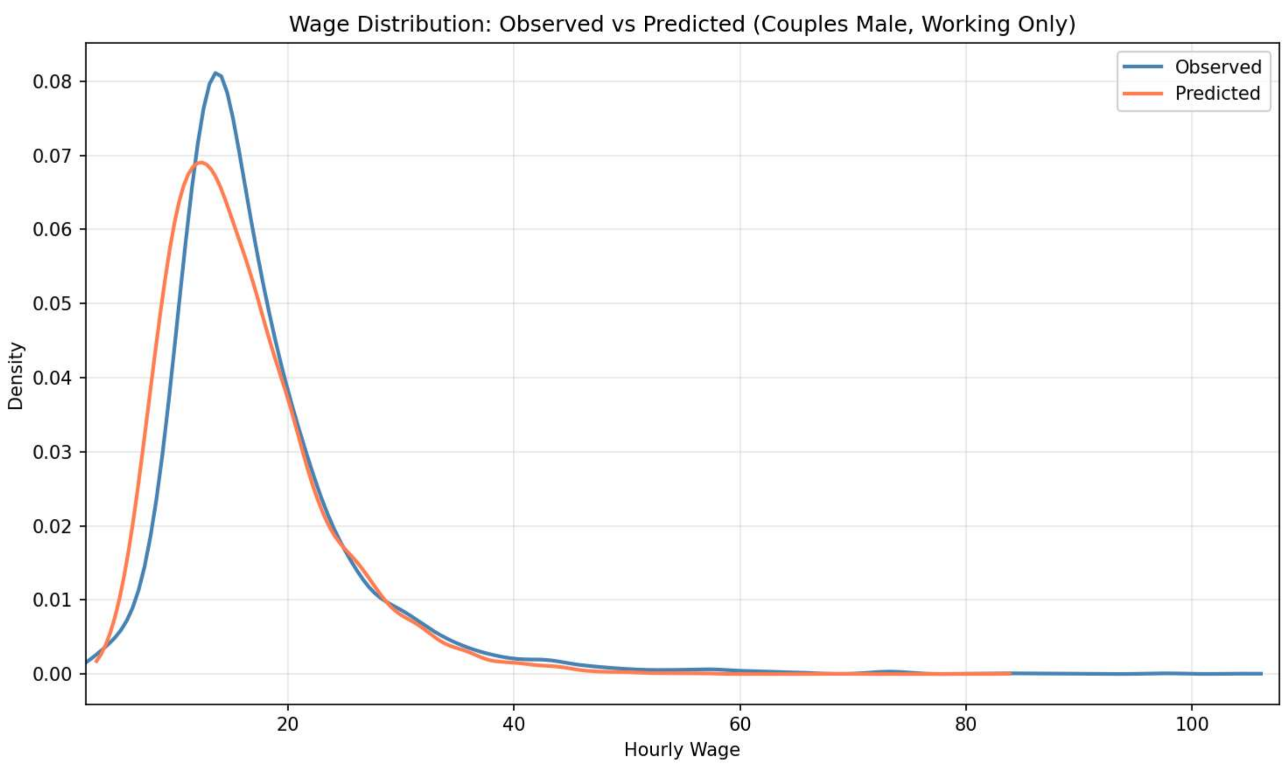


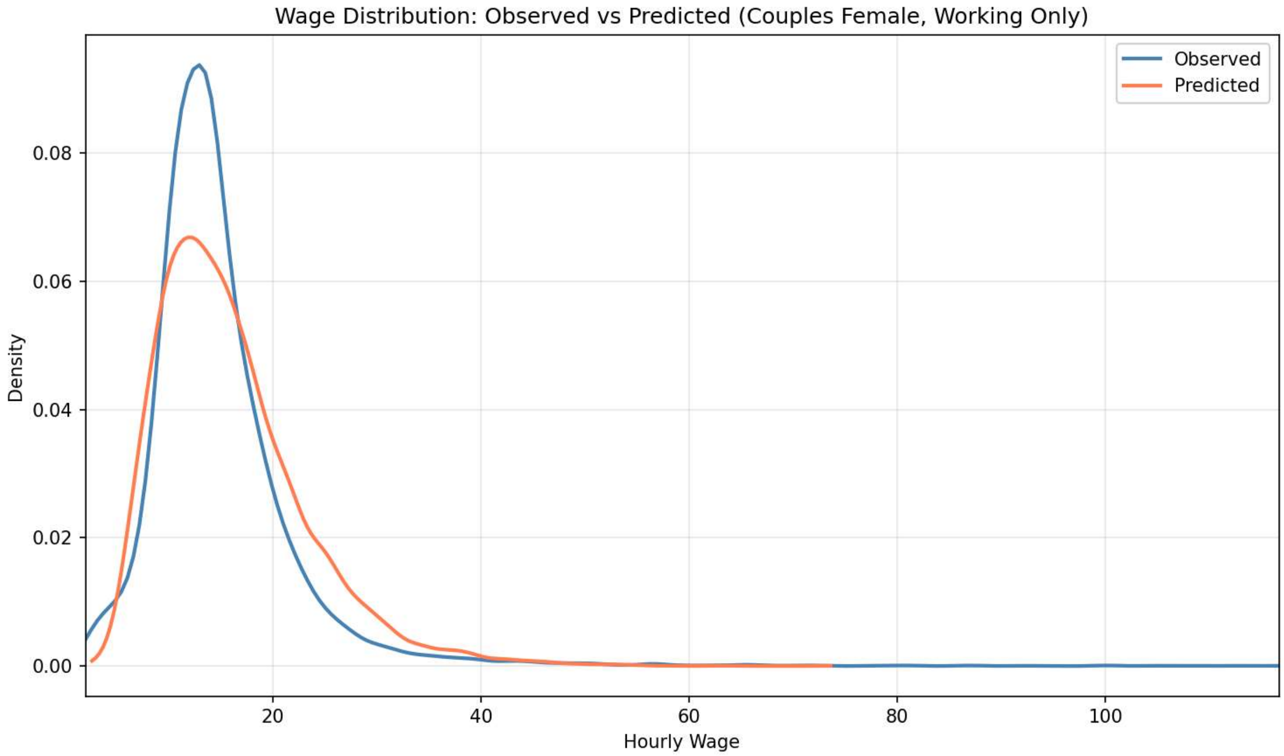
?? Wage Distribution: Observed vs Predicted (Working Only)


Wage Distribution: Observed vs Predicted (Total, Working Only)
Observed
Predicted
Density
0.00
0.02
0.04
0.06
0.08
20
40
60
80
100
Hourly Wage


Wage Distribution: Observed vs Predicted (Singles Male, Working Only)
Observed
Predicted
Density
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
20
40
60
80
100
Hourly Wage


Wage Distribution: Observed vs Predicted (Singles Female, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.02
0.04
0.06
0.08
20
40
60
80


Wage Distribution: Observed vs Predicted (Couples Male, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
20
40
60
80
100



Group-Specific Parameters

Single Males

beta_E_drgmd	-0.6644
beta_E_drgn2	-0.0701
beta_E_drgn3	0.0845
beta_E_drgn4	0.0027
beta_E_drgn5	-0.1415
beta_E_drgn6	-0.0538
beta_E_drgn7	-0.1811
beta_E_drgn8	-0.3444
beta_E_drgur	-0.5323
beta_E_gsur	-1.4041
beta_E_y2015	-0.2510
beta_E_y2017	-0.0642
beta_h_ft	1.0338

Single Females

beta_E_drgmd	-0.6644
beta_E_drgn2	-0.0701
beta_E_drgn3	0.0845
beta_E_drgn4	0.0027
beta_E_drgn5	-0.1415
beta_E_drgn6	-0.0538
beta_E_drgn7	-0.1811
beta_E_drgn8	-0.3444
beta_E_drgur	-0.5323
beta_E_gsur	-1.4041
beta_E_y2015	-0.2510
beta_E_y2017	-0.0642
beta_h_ft	1.0338

beta_h_lh	-1.2443
beta_h_pt1	-1.4275
beta_l0	4.4287
beta_l_age	0.6668
beta_l_age2	0.3699
beta_w0	2.1971
beta_w_educH	0.3384
beta_w_educL	-0.0606
beta_w_pexp	0.3816
beta_w_pexp2	-0.0818
sigma	0.3899
theta_c_singles	0.0095
theta_l	-1.7902

beta_h_lh	-1.2443
beta_h_pt1	-1.4275
beta_l0	3.8481
beta_l_age	0.4902
beta_l_age2	1.0000
beta_l_nkids	1.7818
beta_w0	2.1971
beta_w_educH	0.3384
beta_w_educL	-0.0606
beta_w_pexp	0.3816
beta_w_pexp2	-0.0818
sigma	0.3899
theta_c_singles	0.0095
theta_l	-1.4404

Males in Couples

beta_E	-0.3311
beta_E_drgmd	-0.6644
beta_E_drgn2	-0.0701
beta_E_drgn3	0.0845
beta_E_drgn4	0.0027
beta_E_drgn5	-0.1415
beta_E_drgn6	-0.0538
beta_E_drgn7	-0.1811
beta_E_drgn8	-0.3444
beta_E_drgur	-0.5323
beta_E_gsur	-1.4041
beta_E_y2015	-0.2510
beta_E_y2017	-0.0642
beta_h_ft	1.0338
beta_h_lh	-1.2443
beta_h_pt1	-1.4275
beta_h_pt2	-1.2108
beta_l0	0.0000
beta_l_age	-0.0699

Females in Couples

beta_E	-1.0235
beta_E_drgmd	-0.6644
beta_E_drgn2	-0.0701
beta_E_drgn3	0.0845
beta_E_drgn4	0.0027
beta_E_drgn5	-0.1415
beta_E_drgn6	-0.0538
beta_E_drgn7	-0.1811
beta_E_drgn8	-0.3444
beta_E_drgur	-0.5323
beta_E_gsur	-1.4041
beta_E_y2015	-0.2510
beta_E_y2017	-0.0642
beta_h_ft	1.0338
beta_h_lh	-1.2443
beta_h_pt1	-1.4275
beta_h_pt2	0.3911
beta_l0	10.2593
beta_l_age	-1.9208

beta_l_age2	0.1315
beta_occ_2	-1.6265
beta_occ_3	-2.3279
beta_occ_4	0.2572
beta_w0	2.1971
beta_w_educH	0.3384
beta_w_educL	-0.0606
beta_w_pexp	0.3816
beta_w_pexp2	-0.0818
sigma	0.3899
theta_c_singles	0.0095
theta_l	-0.8000

beta_l_age2	1.0000
beta_l_nkids	0.5212
beta_occ_2	0.0179
beta_occ_3	-0.4041
beta_occ_4	0.8376
beta_w0	2.1971
beta_w_educH	0.3384
beta_w_educL	-0.0606
beta_w_pexp	0.3816
beta_w_pexp2	-0.0818
sigma	0.3899
theta_c_singles	0.0095
theta_l	-2.2273

📋 Parameter Estimates (by block — from DiagnosticsBundle)

Primary SE: Hessian (classical) *(supply --cluster-se-json for cluster-robust SE)*

Primary SE is robust/cluster when --cluster-se-json is supplied; otherwise the Hessian (classical) SE is primary.

Block: preference (19 params)

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_l0_sm	4.428684	—	—	—	—	—				hessian
joint.beta_l_age_sm	0.666821	—	—	—	—	—				hessian
joint.beta_l_age2_sm	0.369904	—	—	—	—	—				hessian
joint.theta_l_sm	-1.790248	—	—	—	—	—				hessian
joint.beta_l0_sf	3.848124	—	—	—	—	—				hessian
joint.beta_l_age_sf	0.490208	—	—	—	—	—				hessian
joint.beta_l_age2_sf	1.000000	—	—	—	—	—			⬆	hessian
joint.beta_l_nkids_sf	1.781787	—	—	—	—	—				hessian
joint.theta_l_sf	-1.440449	—	—	—	—	—				hessian
joint.theta_c_singles	0.009507	—	—	—	—	—				hessian
joint.beta_l0_m	1.000000e-06	—	—	—	—	—		⬇		hessian

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_l_age_m	-0.069861	—	—	—	—	—				hessian
joint.beta_l_age2_m	0.131498	—	—	—	—	—				hessian
joint.beta_l0_f	10.259291	—	—	—	—	—				hessian
joint.beta_l_age_f	-1.920809	—	—	—	—	—				hessian
joint.beta_l_age2_f	1.000000	—	—	—	—	—			↑	hessian
joint.beta_l_nkids_f	0.521224	—	—	—	—	—				hessian
joint.theta_l_f	-2.227295	—	—	—	—	—				hessian
joint.theta_l_m	-0.800000	—	—	—	—	—				hessian

Block: employment_hours_opportunity (19 params)

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_E_m	-0.331053	—	—	—	—	—				hessian
joint.beta_E_f	-1.023511	—	—	—	—	—				hessian
joint.beta_h_pt1	-1.427496	—	—	—	—	—				hessian
joint.beta_h_pt2_m	-1.210799	—	—	—	—	—				hessian
joint.beta_h_pt2_f	0.391147	—	—	—	—	—				hessian
joint.beta_h_ft	1.033849	—	—	—	—	—				hessian
joint.beta_h_lh	-1.244271	—	—	—	—	—				hessian
joint.beta_E_gsur	-1.404148	—	—	—	—	—				hessian
joint.beta_E_drgn2	-0.070089	—	—	—	—	—				hessian
joint.beta_E_drgn3	0.084490	—	—	—	—	—				hessian
joint.beta_E_drgn4	0.002706	—	—	—	—	—				hessian
joint.beta_E_drgn5	-0.141500	—	—	—	—	—				hessian
joint.beta_E_drgn6	-0.053847	—	—	—	—	—				hessian
joint.beta_E_drgn7	-0.181117	—	—	—	—	—				hessian
joint.beta_E_drgn8	-0.344429	—	—	—	—	—				hessian
joint.beta_E_y2015	-0.250966	—	—	—	—	—				hessian
joint.beta_E_y2017	-0.064186	—	—	—	—	—				hessian
joint.beta_E_drgur	-0.532338	—	—	—	—	—				hessian
joint.beta_E_drgmd	-0.664402	—	—	—	—	—				hessian

Block: wage_opportunity (6 params)

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_w0	2.197091	—	—	—	—	—				hessian
joint.beta_w_educL	-0.060629	—	—	—	—	—				hessian
joint.beta_w_educH	0.338371	—	—	—	—	—				hessian

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_w_pexp	0.381648	—	—	—	—	—				hessian
joint.beta_w_pexp2	-0.081830	—	—	—	—	—				hessian
joint.sigma	0.389943	—	—	—	—	—				hessian

Block: occupation_opportunity (6 params)

parameter	estimate	se_hessian	t_hessian	se_robust	t_robust	p_robust	fixed	at_lower	at_upper	primary_se
joint.beta_occ_2_m	-1.626456	—	—	—	—	—				hessian
joint.beta_occ_3_m	-2.327864	—	—	—	—	—				hessian
joint.beta_occ_4_m	0.257237	—	—	—	—	—				hessian
joint.beta_occ_2_f	0.017888	—	—	—	—	—				hessian
joint.beta_occ_3_f	-0.404136	—	—	—	—	—				hessian
joint.beta_occ_4_f	0.837571	—	—	—	—	—				hessian

📋 Parameter Estimates by Category (legacy view)

*Significance: *** p<0.001, ** p<0.01, * p<0.05. The bundle-driven block view above is the canonical Phase-2 source.*

☐ Bounded (has constraints) ☐ ⚠ Hit bound ☐ Warning: degenerate SE (near zero) ☐ p ∈ [0.05, 0.1) ☐ p ∈ [0.1, 0.25) ☐ p ≥ 0.25

🎯 Preference Parameters (Utility Function)

Parameters determining the utility from consumption (β_c, θ_c) and leisure (β_l, θ_l) by demographic group.*

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l0_sm	4.4287	N/A	N/A	N/A	0.0500	50.0000	1.0000
joint.beta_l_age_sm	0.6668	N/A	N/A	N/A	-5.0000	5.0000	0.0000
joint.beta_l_age2_sm	0.3699	N/A	N/A	N/A	-1.0000	1.0000	0.0000
joint.theta_l_sm	-1.7902	N/A	N/A	N/A	-8.0000	0.9500	-1.0000
joint.beta_l0_sf	3.8481	N/A	N/A	N/A	0.0500	50.0000	1.0000
joint.beta_l_age_sf	0.4902	N/A	N/A	N/A	-5.0000	5.0000	0.0000
joint.beta_l_age2_sf	**1.0000**	**N/A**	**N/A**	**N/A**	**-1.0000**	**1.0000**	**0.0000**
joint.beta_l_nkids_sf	1.7818	N/A	N/A	N/A	-5.0000	5.0000	0.0000
joint.theta_l_sf	-1.4404	N/A	N/A	N/A	-8.0000	0.9500	-1.0000

⏰ Employment and Hours Opportunity Parameters

Working-gated employment intercept and hours focal-point indicators, plus residual market-access shifters (gsur, education) declared in `market_opportunity.shifters`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_h_pt1	-1.4275	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_h_pt2_m	-1.2108	N/A	N/A	N/A	-10.0000	10.0000	-1.1900
joint.beta_h_pt2_f	0.3911	N/A	N/A	N/A	-10.0000	10.0000	0.3700
joint.beta_h_ft	1.0338	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_h_lh	-1.2443	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_gsur	-1.4041	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn2	-0.0701	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn3	0.0845	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn4	0.0027	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn5	-0.1415	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn6	-0.0538	N/A	N/A	N/A	-10.0000	10.0000	0.0000
joint.beta_E_drgn7	-0.1811	N/A	N/A	N/A	-10.0000	10.0000	0.0000

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\mu_w(X)$ intercept, education, experience, and residual standard deviation σ.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_w0	2.1971	N/A	N/A	N/A	-10.0000	20.0000	2.0000
joint.beta_w_educL	-0.0606	N/A	N/A	N/A	-5.0000	5.0000	-0.1000
joint.beta_w_educH	0.3384	N/A	N/A	N/A	-5.0000	5.0000	0.2000
joint.beta_w_pexp	0.3816	N/A	N/A	N/A	-1.0000	1.0000	0.0200
joint.beta_w_pexp2	-0.0818	N/A	N/A	N/A	-0.1000	0.1000	-0.0003
joint.sigma	0.3899	N/A	N/A	N/A	0.1000	20.0000	0.5000

🧭 Occupation Opportunity Parameters

Working-gated occupation shifters declared in `occupation_opportunity.shifters`; routed by `applies_to` group (sm/sf/cm/cf). Reference category is the YAML's `occupation_opportunity.reference`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_2_m	-1.6265	N/A	N/A	N/A	-15.0000	15.0000	0.0000
joint.beta_occ_3_m	-2.3279	N/A	N/A	N/A	-15.0000	15.0000	0.0000
joint.beta_occ_4_m	0.2572	N/A	N/A	N/A	-15.0000	15.0000	0.0000
joint.beta_occ_2_f	0.0179	N/A	N/A	N/A	-15.0000	15.0000	0.0000

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_3_f	-0.4041	N/A	N/A	N/A	-15.0000	15.0000	0.0000
joint.beta_occ_4_f	0.8376	N/A	N/A	N/A	-15.0000	15.0000	0.0000

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